

ML – Lab

Learning Types:

1. Supervised - Labelled - Predicts OP with given IP via labels
2. Unsupervised - Un labelled - Deduces Pattern to answer to IP
3. Reinforcement Learning - It explores all outcomes and when expected outcome comes you give it a positive feedback and when it's not correct give it a negative score. It keeps on moving to positive score to learn and Answer.

Key Topics:

1. Regression Models
2. Classification Models
3. Clustering Models
4. Evaluation Metrics
 1. Accuracy
 2. Precision
 3. Recall
 4. F1 Score
 5. Specificity (opposite of precision)
5. Dimensionality Reduction & Ensemble learning

- **Dimensionality Reduction:** These techniques aim to reduce the number of input variables while preserving important information.
- **Ensemble Learning:** Ensemble learning combines multiple ML models to make predictions or decisions. It improves accuracy, reduces overfitting, and includes techniques like bagging, boosting, and stacking.

